PROJECT DOCUMENTATION

ETL Process for Medicare Inpatient Data 2017

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BY: RITESH CHAUDHARY

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1. Abstract

This ETL documentation report presents a comprehensive, end-to-end data engineering workflow designed to extract, transform, and load healthcare-related inpatient records into a structured analytical data warehouse. The system follows a multi-layer architecture utilizing raw staging, cleaned transformation tables, star-schema modeling, and optimized SQL-based workflows.

Visual ER diagrams, data flow models, and SQL scripts are included to illustrate the technical foundation. The project incorporates business analytical queries that support decision-making.

2. Introduction

Modern healthcare institutions generate large volumes of clinical and administrative data. This project focuses on developing an ETL pipeline that processes patient hospitalization data into meaningful analytical insights. SQL Server is used as the main RDBMS, with structured ingestion from CSV files into a multi-layer ETL environment.

The system analyzes DRG categories, financial metrics, doctor performance, mortality, demographic trends, and collection reports.

3. Project Scope

The ETL project includes the following:

- Extracting structured CSV files containing inpatient records.
- Designing logical and physical schemas.
- Bulk loading data from external sources.
- Cleansing and transforming data.
- Building a star schema for analytical processing.
- Developing 10 analytical queries.
- Implementing data quality checks.
- Documenting all technical steps.

4. About Source System & Data Description

The source dataset originates from hospital inpatient records containing fields such as DRG code, patient name, district, doctor, admission/discharge dates, gender, age, financials, and treatment outcomes. The source format is a delimited CSV file, uploaded as part of the ETL ingestion pipeline.

5. ER Diagram

The following ER model represents the logical relationship between dimension tables like patients, doctors, districts, DRG details, and hospitalization facts.

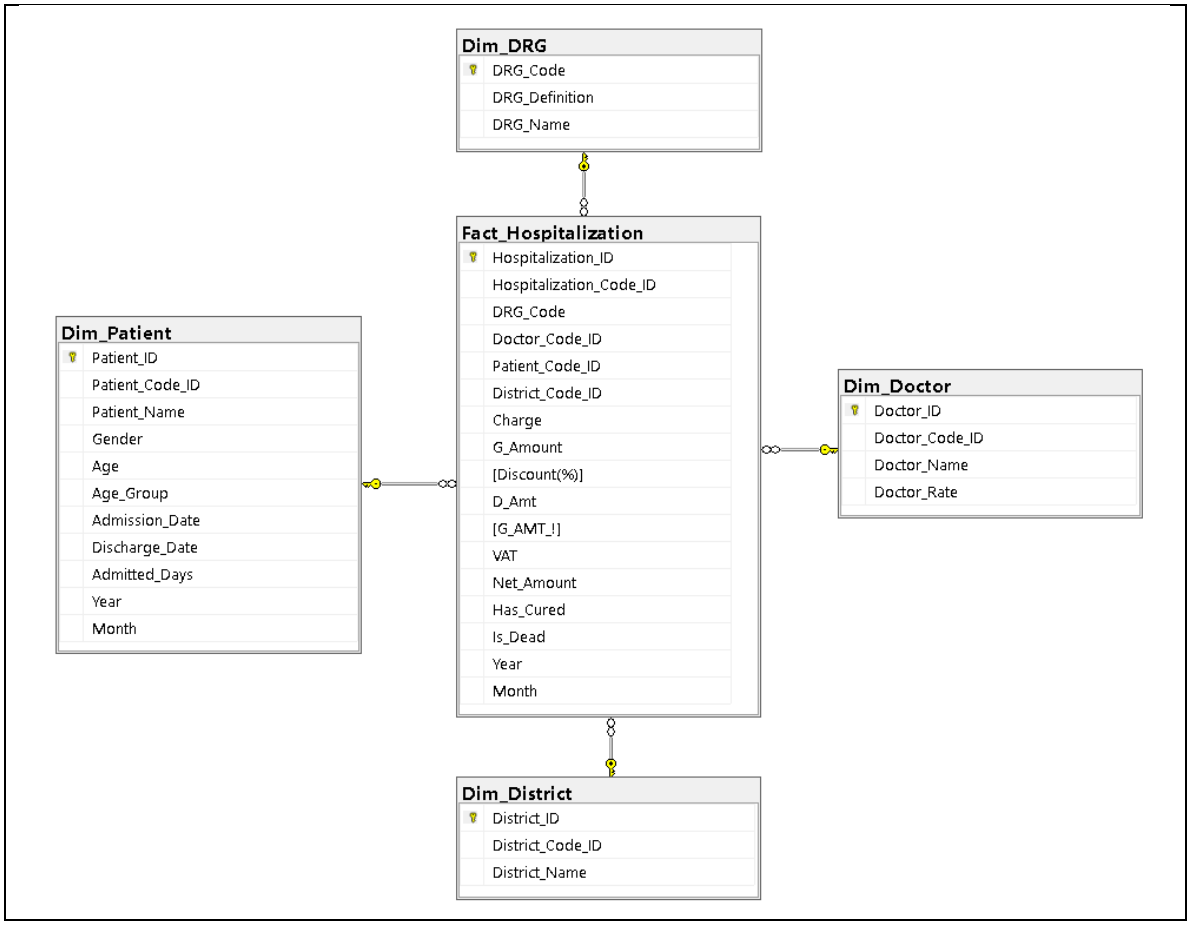


Fig 1: *ER Diagram (Star Schema)*

6. ETL Architecture Overview

A multi-layer ETL structure is used, including:

- **Source Layer** – Raw CSV files.
- **Staging Layer (Raw Schema)** – Temporary tables for storing unprocessed data.
- **Transform Layer (Clean Schema)** – Standardized and validated tables.
- **Data Warehouse Layer (DWH Schema)** – Star Schema (Fact + Dimensions)
- **Consume Layer** – For Business Insights (Reporting)

7. Data Flow Diagram

The data flow diagram (DFD) outlines how the data moves through extraction, transformation, and loading until delivered to analytics.

The DFD outlines these steps:

1. CSV File Extraction
2. Bulk Load into Raw Schema
3. Cleansing & Transformations & Load into Clean Schema
4. Loading into Fact and Dimension tables with keys mapping
5. 10 Business Queries for Reporting

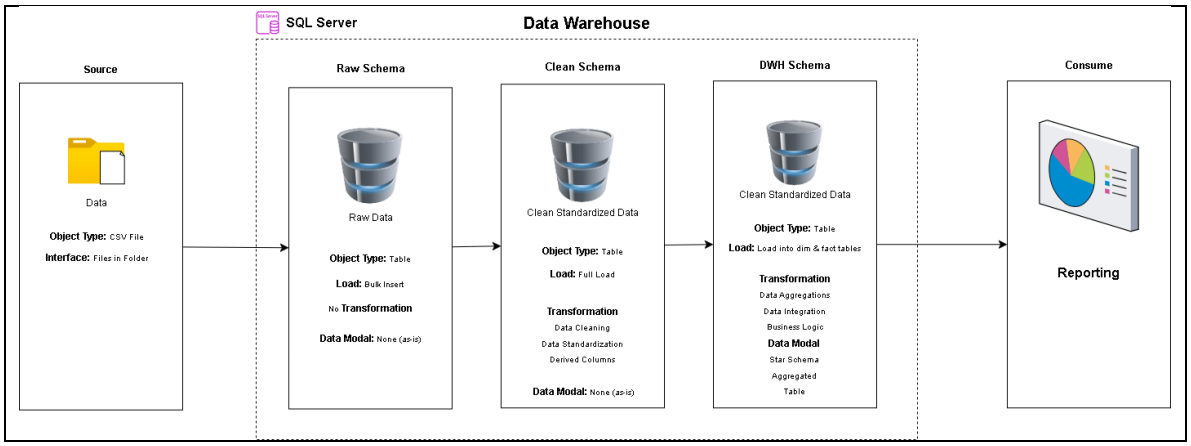


Fig 2: *DFD (Data Flow Diagram)*

8. Schema Structure & Rationale

Three schemas were used purposely:

Schema	Purpose
Raw Schema	Holds unmodified bulk-loaded data
Clean Schema	Stores cleaned, validated, and standardized data
DWH Schema	Contains star schema tables for analytics

Advantages of Schema Separation

- Layer-wise separation improves maintainability
- Ensure traceability from source, staging and final tables.
- Prevents contamination of analytical data.
- Allows independent auditing, rollback, and quality checks.

9. Bulk Insert Strategy

High-volume ingestion uses SQL Server's BULK INSERT mechanism.

```
-- bulk insert from .csv file
BULK INSERT Staging_Raw
FROM 'C:\Users\WINDOWS10\Downloads\Medicare-Inpatient-2017.csv'
WITH (
    FIRSTROW = 2,
    FIELDTERMINATOR = '|',
    ROWTERMINATOR = '\n'
);
/*
(196325 rows affected)
.xlxs file: 196326 (1 is header)
*/
```

Fig 3: *Bulk insert*

The objective is high-speed ingestion with minimal overhead.

10. Raw Layer (Staging) Processing

Raw data is inserted without modification. This stage helps:

- Capture original data for audit
- Detect Anomalies
- Prevent mixing raw and processed data
- Enable repeatability during pipeline reruns

Raw Staging Table Structure:

```
CREATE TABLE Staging_Raw (
    DRG_Code VARCHAR(30),
    DRG_Definition VARCHAR(100),
    Patient_Name VARCHAR(50),
    District VARCHAR(50),
    Doctor VARCHAR(50),
    Admission_Date VARCHAR(50),
    Discharge_date VARCHAR(50),
    Admitted_Days VARCHAR(50),
    Gender VARCHAR(10),
    Age VARCHAR(10),
    Charge VARCHAR(25),
    G_Amount VARCHAR(25),
    "Discount(%)" VARCHAR(25),
    D_Amt VARCHAR(25),
    "G_AMT_!" VARCHAR(25),
    VAT VARCHAR(25),
    Net_Amount VARCHAR(25),
    Has_Cured VARCHAR(10),
    IS_DEAD VARCHAR(10),
    Year VARCHAR(25),
    month VARCHAR(25),
    Doctor_Rate VARCHAR(25)
);
```

Fig 4: *Raw Staging Table Structure*

At this stage every data is stored from ‘Bulk Insert” in text format. Later data are transformed into different data types.

11. Clean/Transformed Layer Processing

The clean layer corrects formatting issues and ensures data consistency.

Clean Staging Table Structure:

```
CREATE TABLE Staging_Clean (
    DRG_Code INT,
    DRG_Definition VARCHAR(100),
    DRG_Name VARCHAR(100),
    Patient_Name VARCHAR(50),
    District VARCHAR(50),
    Doctor VARCHAR(50),
    Admission_Date DATE,
    Discharge_Date DATE,
    Admitted_Days INT,
    Gender VARCHAR(10),
    Age INT,
    Charge INT,
    G_Amount INT,
    "Discount(%)" DECIMAL(4, 2),
    D_Amt DECIMAL(20, 5),
    "G_AMT_!" DECIMAL(20, 5),
    VAT DECIMAL(20, 5),
    Net_Amount DECIMAL(20, 5),
    Has_Cured VARCHAR(10),
    Is_Dead VARCHAR(10),
    Year INT,
    Month INT,
    Doctor_Rate DECIMAL(2,1)
);
```

Fig 5: *Clean Staging Table Structure*

Clean Data Insert:

```
INSERT INTO Staging_Clean
(DRG_Code, DRG_Definition, DRG_Name, Patient_Name, District, Doctor, Admission_Date, Discharge_Date,
Admitted_Days, Gender, Age, Charge, G_Amount, [Discount(%)], D_Amt, [G_AMT_!], VAT, Net_Amount,
 Has_Cured, Is_Dead, Year, Month, Doctor_Rate)
SELECT
    CAST(NULLIF(DRG_Code, '') AS INT),
    NULLIF(DRG_Definition, ''),
    LTRIM(SUBSTRING(
        DRG_Definition,
        CHARINDEX('-', DRG_Definition) + 1,
        LEN(DRG_Definition)
    )),
    NULLIF(Patient_Name, ''),
    CAST(NULLIF(District, '') AS VARCHAR(25)),
    CAST(NULLIF(Doctor, '') AS VARCHAR(25)),
    CONVERT(DATE, NULLIF(Admission_Date, ''), 101),
    CONVERT(DATE, NULLIF(Discharge_Date, ''), 101),
    CAST(NULLIF(Admitted_Days, '') AS INT),
    CAST(NULLIF(Gender, '') AS VARCHAR(10)),
    CAST(NULLIF(Age, '') AS INT),
    CAST(NULLIF(Charge, '') AS INT),
    CAST(NULLIF(G_Amount, '') AS INT),
    CAST(NULLIF([Discount(%)], '') AS DECIMAL(4,2)),
    CAST(NULLIF(D_Amt, '') AS DECIMAL(20,5)),
    CAST(NULLIF([G_AMT_!], '') AS DECIMAL(20,5)),
    CAST(NULLIF(VAT, '') AS DECIMAL(20,5)),
    CAST(NULLIF(Net_Amount, '') AS DECIMAL(20,5)),
    CAST(NULLIF(Has_Cured, '') AS VARCHAR(5)),
    CAST(NULLIF(IS_DEAD, '') AS VARCHAR(5)),
    CAST(NULLIF(Year, '') AS INT),
    CAST(NULLIF(Month, '') AS INT),
    CAST(NULLIF(Doctor_Rate, '') AS DECIMAL(2,1))
FROM Staging_Raw; -- 196325
```

Fig 6: *Clean Data Insert*

Transformations performed:

- Trimming text
- Converting data types
- Handling nulls
- Ensuring valid dates

12. Fact & Dimension Table Design (Star Schema)

The Star schema enhances analytical performance.

Dimension Tables

District

Table structure and insert data

```
CREATE TABLE Dim_District (
    District_ID INT IDENTITY(1,1) PRIMARY KEY,
    District_Code_ID AS (CAST('DIS-' + CAST(District_ID AS VARCHAR(20)) AS VARCHAR(20))) PERSISTED UNIQUE,
    District_Name VARCHAR(50)
);
-- insert
INSERT INTO Dim_District (District_Name)
SELECT DISTINCT District
FROM Staging_Clean; -- 59
```

Fig 7: *Dimension District Table Structure & Insert Distinct Data*

Doctor

Table structure and insert data

```
CREATE TABLE Dim_Doctor (
    Doctor_ID INT IDENTITY(1,1) PRIMARY KEY,
    Doctor_Code_ID AS (CAST('DOC-' + CAST(Doctor_ID AS VARCHAR(20)) AS VARCHAR(20))) PERSISTED UNIQUE,
    Doctor_Name VARCHAR(50),
    Doctor_Rate DECIMAL(2,1)
);
-- insert
INSERT INTO Dim_Doctor (Doctor_Name, Doctor_Rate)
SELECT DISTINCT Doctor, Doctor_Rate
FROM Staging_Clean; -- 29
```

Fig 8: *Dimension Doctor Table Structure & Insert Distinct Data*

DRG (Diagnosis Related Group)

Table structure and insert data

```sql
CREATE TABLE Dim_DRG (
    DRG_Code INT PRIMARY KEY,
    DRG_Definition VARCHAR(100),
    DRG_Name VARCHAR(100)
);
-- insert
INSERT INTO Dim_DRG (DRG_Code, DRG_Definition, DRG_Name)
SELECT DISTINCT DRG_Code, DRG_Definition, DRG_Name
FROM Staging_Clean; -- 563
```

Fig 9: *Dimension DRG Table Structure & Insert Distinct Data*

Patient

Table structure and insert data

```sql
CREATE TABLE Dim_Patient (
    Patient_ID INT IDENTITY (1,1) PRIMARY KEY,
    Patient_Code_ID AS (CAST('PAT-' + CAST(Patient_ID AS VARCHAR(20)) AS VARCHAR(20))) PERSISTED UNIQUE,
    Patient_Name VARCHAR(50),
    Gender VARCHAR(10),
    Age INT,
    Age_Group VARCHAR(50),
    Admission_Date DATE,
    Discharge_Date DATE,
    Admitted_Days INT,
    Year INT,
    Month INT
);
-- Insert UNIQUE patients (using DISTINCT)
INSERT INTO Dim_Patient (Patient_Name, Gender, Age, Age_Group, Admission_Date, Discharge_Date, Admitted_Days, Year, Month)
SELECT DISTINCT
    Patient_Name,
    Gender,
    Age,
    CASE
        WHEN Age < 1 THEN 'Infant'
        WHEN Age BETWEEN 1 AND 12 THEN 'Child'
        WHEN Age BETWEEN 13 AND 19 THEN 'Teen'
        WHEN Age BETWEEN 20 AND 39 THEN 'Adult'
        WHEN Age BETWEEN 40 AND 59 THEN 'Middle Age'
        WHEN Age >= 60 THEN 'Senior'
        ELSE 'Unknown'
    END AS Age_Group,
    Admission_Date,
    Discharge_Date,
    Admitted_Days,
    Year,
    Month
FROM Staging_Clean; -- 196303
```

Fig 10: *Dimension Patient Table Structure & Insert Distinct Data*

Fact Table

Stores measures such as charges, VAT, Net amount, etc.

Hospitalization

Table structure and insert data

```
CREATE TABLE Fact_Hospitalization (
    Hospitalization_ID INT IDENTITY(1,1) PRIMARY KEY,
    Hospitalization_Code_ID AS (CAST('HOS-' + CAST(Hospitalization_ID AS VARCHAR(20)) AS VARCHAR(20))) PERSISTED UNIQUE,
    DRG_Code INT,
    Doctor_Code_ID VARCHAR(20),
    Patient_Code_ID VARCHAR(20),
    District_Code_ID VARCHAR(20),
    Charge INT,
    G_Amount INT,
    "Discount(%)" DECIMAL(4, 2),
    D_Amt DECIMAL(20, 5),
    "G_AMT_!" DECIMAL(20, 5),
    VAT DECIMAL(20, 5),
    Net_Amount DECIMAL(20, 5),
    Has_Cured VARCHAR(10),
    Is_Dead VARCHAR(10),
    Year INT,
    Month INT,
    FOREIGN KEY (DRG_Code) REFERENCES Dim_DRG(DRG_Code),
    FOREIGN KEY (Doctor_Code_ID) REFERENCES Dim_Doctor(Doctor_Code_ID),
    FOREIGN KEY (Patient_Code_ID) REFERENCES Dim_Patient(Patient_Code_ID),
    FOREIGN KEY (District_Code_ID) REFERENCES Dim_District(District_Code_ID)
);
-- insert
INSERT INTO Fact_Hospitalization (Charge, G_Amount, [Discount(%)], D_Amt, [G_AMT_!], VAT, Net_Amount,
Has_Cured, Is_Dead, Year, Month, DRG_Code, Doctor_Code_ID, Patient_Code_ID, District_Code_ID)
SELECT DISTINCT
    m.Charge, m.G_Amount, m.[Discount(%)], m.D_Amt, m.[G_AMT_!], m.VAT, m.Net_Amount,
    m.Has_Cured, m.Is_Dead, m.Year, m.Month, drg.DRG_Code, doc.Doctor_Code_ID, p.Patient_Code_ID, d.District_Code_ID
FROM Staging_Clean m
INNER JOIN Dim_DRG drg ON m.DRG_Definition = drg.DRG_Definition AND m.DRG_Name = drg.DRG_Name
INNER JOIN Dim_Doctor doc ON m.Doctor = doc.Doctor_Name AND m.Doctor_Rate = doc.Doctor_Rate
INNER JOIN Dim_Patient p ON m.Patient_Name = p.Patient_Name AND m.Gender = p.Gender AND m.Age = p.Age
    AND m.Admission_Date = p.Admission_Date AND m.Discharge_Date = p.Discharge_Date AND m.Admitted_Days = p.Admitted_Days
    AND m.Year = p.Year AND m.Month = p.Month
INNER JOIN Dim_District d ON m.District = d.District_Name; -- 196325
```

Fig 11: *Fact Hospitalization Table Structure & Insert Distinct Data*

13. Business Analytical Queries and PowerBI Visual with Explanation

Below are the 10 business analytical SQL queries used in this ETL project. Each query is included exactly as provided in the project and described in paragraph form.

Query 1: Which DRG treatments generate the highest revenue?

Question: Which top 20 DRG treatment categories generate the highest total revenue, and how do they rank against other treatments?

```sql
/*
1. Which DRG treatments generate the highest revenue?
Question: Which top 20 DRG treatment categories generate the highest total revenue,
    and how do they rank against other treatments?
*/
SELECT TOP 20
    drg.DRG_Code,
    drg.DRG_Name,
    COUNT(*) AS Total_Cases,
    SUM(CAST(f.Net_Amount AS DECIMAL(18,2))) AS Total_Revenue,
    AVG(CAST(f.Net_Amount AS DECIMAL(18,2))) AS Avg_Revenue_Per_Case
FROM Fact_Hospitalization f
INNER JOIN Dim_DRG drg
    ON f.DRG_Code = drg.DRG_Code
GROUP BY
    drg.DRG_Code,
    drg.DRG_Name
ORDER BY
    Total_Revenue DESC;
```

Results | Messages

	DRG_Code	DRG_Name	Total_Cases	Total_Revenue	Avg_Revenue_Per_Case
1	871	SEPTICEMIA OR SEVERE SEPSIS W/O MV >96 HOURS W MCC	2838	37249469.87	13125.253653
2	291	HEART FAILURE & SHOCK W MCC	2742	36051982.78	13148.060824
3	190	CHRONIC OBSTRUCTIVE PULMONARY DISEASE W MCC	2687	34559046.27	12861.572858
4	470	MAJOR JOINT REPLACEMENT OR REATTACHMENT OF LOWER ...	2666	34530499.91	12952.175510
5	872	SEPTICEMIA OR SEVERE SEPSIS W/O MV >96 HOURS W/O MCC	2632	33978473.81	12909.754487
6	392	ESOPHAGITIS, GASTROENT & MISC DIGEST DISORDERS W/O ...	2586	33099009.61	12799.307660
7	690	KIDNEY & URINARY TRACT INFECTIONS W/O MCC	2584	33079005.21	12801.472604
8	194	SIMPLE PNEUMONIA & PLEURISY W CC	2517	32339545.58	12848.448780
9	189	PULMONARY EDEMA & RESPIRATORY FAILURE	2465	32134478.82	13036.299724
10	683	RENAL FAILURE W CC	2426	31325271.93	12912.313244
11	603	CELLULITIS W/O MCC	2464	30877636.34	12531.508254

Fig 11: *Top 20 DRG Treatments by Revenue Query*

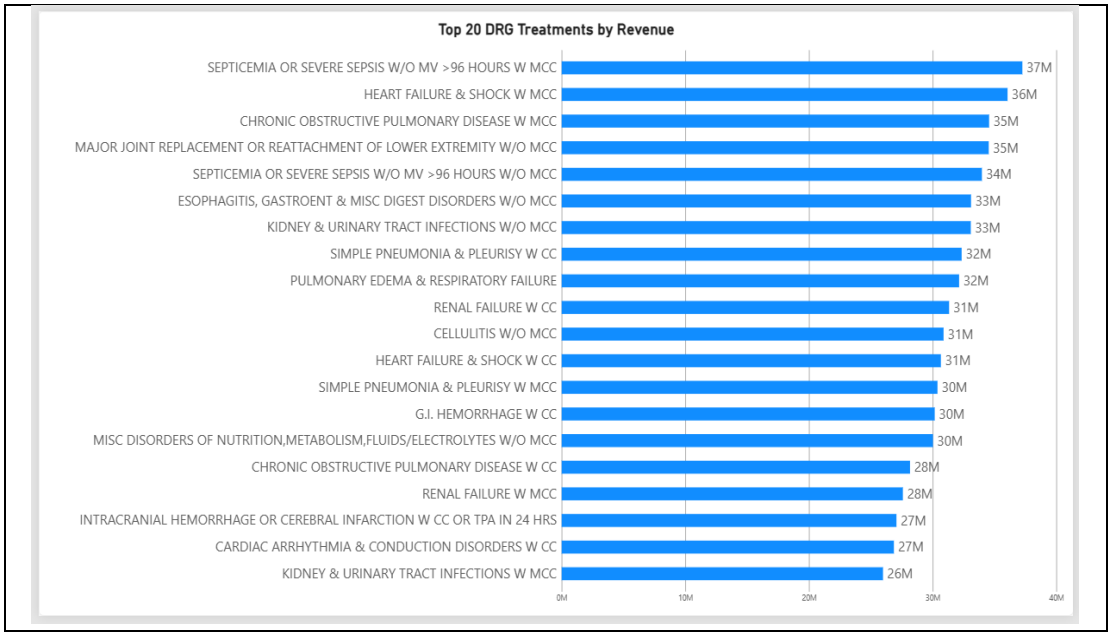


Fig 12: *Top 20 DRG Treatments by Revenue Visual*

Business Value:

Management can identify the highest-revenue treatment categories and use the result for resource allocation, capacity planning and service strategy.

Query 2: Which districts are the strongest markets for the hospital?

Question: Which geographic districts contribute the most patients and revenue, and which districts may represent growth opportunities?

```
/*
2. Which districts are the strongest markets for the hospital?
Question: Which geographic districts contribute the most patients and revenue,
    and which districts may represent growth opportunities?
*/
SELECT TOP 10
    d.District_Name,
    COUNT(*) AS Total_Hospitalizations,
    SUM(CAST(f.Net_Amount AS DECIMAL(18,2))) AS Total_Revenue,
    AVG(CAST(f.Net_Amount AS DECIMAL(18,2))) AS Avg_Revenue_Per_Case
FROM Fact_Hospitalization f
INNER JOIN Dim_District d
    ON f.District_Code_ID = d.District_Code_ID
GROUP BY
    d.District_Name
ORDER BY
    Total_Revenue DESC;
/*
```

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Results Messages

	District_Name	Total_Hospitalizations	Total_Revenue	Avg_Revenue_Per_Case
1	Oregon	3882	51706763.99	13319.619781
2	Idaho	3881	50424306.07	12992.606562
3	Montana	3882	50416196.57	12987.170677
4	Rhodelsland	3882	50415910.33	12987.096942
5	Iowa	3882	50404042.09	12984.039693
6	Hawaii	3882	50358147.96	12972.217403
7	NorthCarolina	3881	50323883.52	12966.731131
8	Missouri	3881	50317999.15	12965.214931
9	Indiana	3881	50273718.36	12953.805297
10	SouthDakota	3882	50264653.50	12948.133307

Fig 14: *Top 10 District by Hospital Revenue Query*

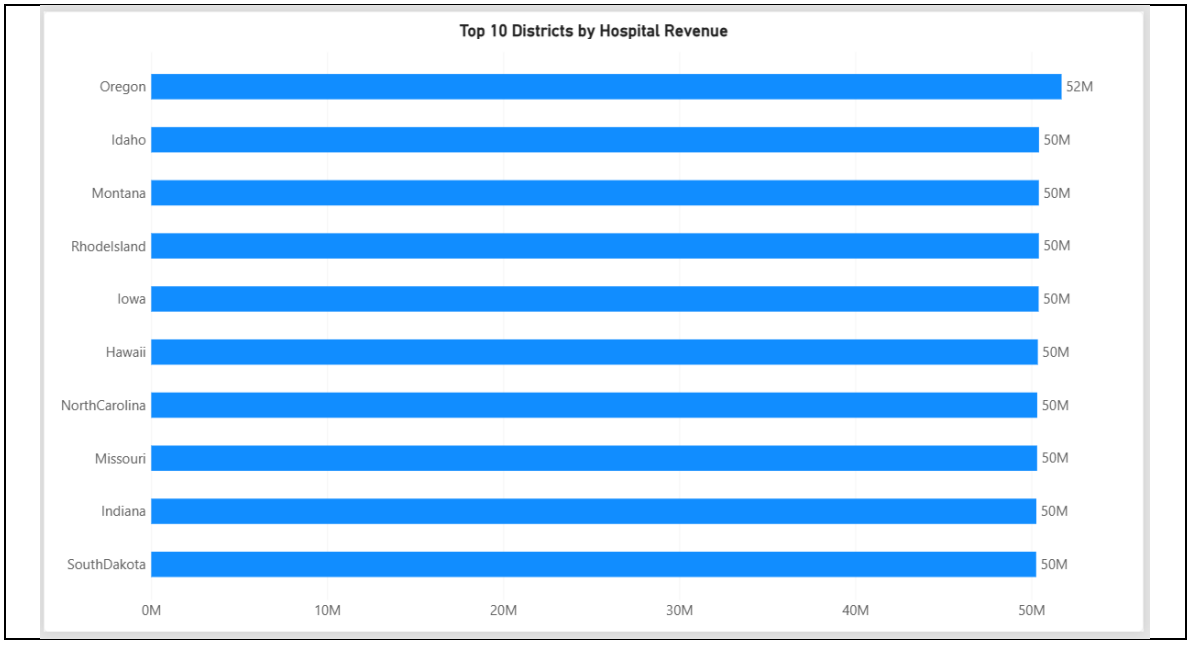


Fig 15: *Top 10 District by Hospital Visual*

Business Value: This can support regional marketing, referral partnerships and service expansion decisions.

Query 3: Which doctors contribute the most revenue while handling significant case volumes?

Question: Which doctors generate the highest revenue while managing a significant number of hospitalizations?

```sql
/*
3. Which doctors contribute the most revenue while handling significant case volumes?
Which doctors generate the highest revenue while managing a significant number of hospitalizations?
*/
SELECT
    doc.Doctor_Name,
    doc.Doctor_Rate,
    COUNT(*) AS Total_Cases,
    SUM(CAST(f.Net_Amount AS DECIMAL(18,2))) AS Total_Revenue,
    AVG(CAST(f.Net_Amount AS DECIMAL(18,2))) AS Avg_Revenue_Per_Case
FROM Fact_Hospitalization f
INNER JOIN Dim_Doctor doc
    ON f.Doctor_Code_ID = doc.Doctor_Code_ID
GROUP BY
    doc.Doctor_Name,
    doc.Doctor_Rate
HAVING COUNT(*) >= 100;
```

98 % 5 0

Results Messages

	Doctor_Name	Doctor_Rate	Total_Cases	Total_Revenue	Avg_Revenue_Per_Case
1	Anish	0.0	12566	159955280.82	12729.212225
2	Anuja	0.0	991	12951881.63	13069.507194
3	Binita	0.0	12375	159415685.61	12882.075604
4	Hari	0.0	12476	160473144.33	12862.547637
5	Javed	0.0	1208	15534685.01	12859.838584
6	Liza	0.0	12424	159709702.85	12854.934228
7	Madan	0.0	1200	15733526.73	13111.272275
8	Malik	0.0	134	1869831.98	13953.970000
9	Manik	0.0	12375	158488698.87	12807.167585
10	Manish	0.0	12928	165996497.82	12840.075635
11	Nabin	0.0	12790	165727125.49	12957.554768
12	Nabina	0.0	972	12343613.04	12699.190370
13	Praful	0.0	12376	156505581.25	12645.893766
14	Rajeev	0.0	12439	160242026.74	12882.227408

Fig 16: *Doctor Revenue vs Hospitalization Volume Query*

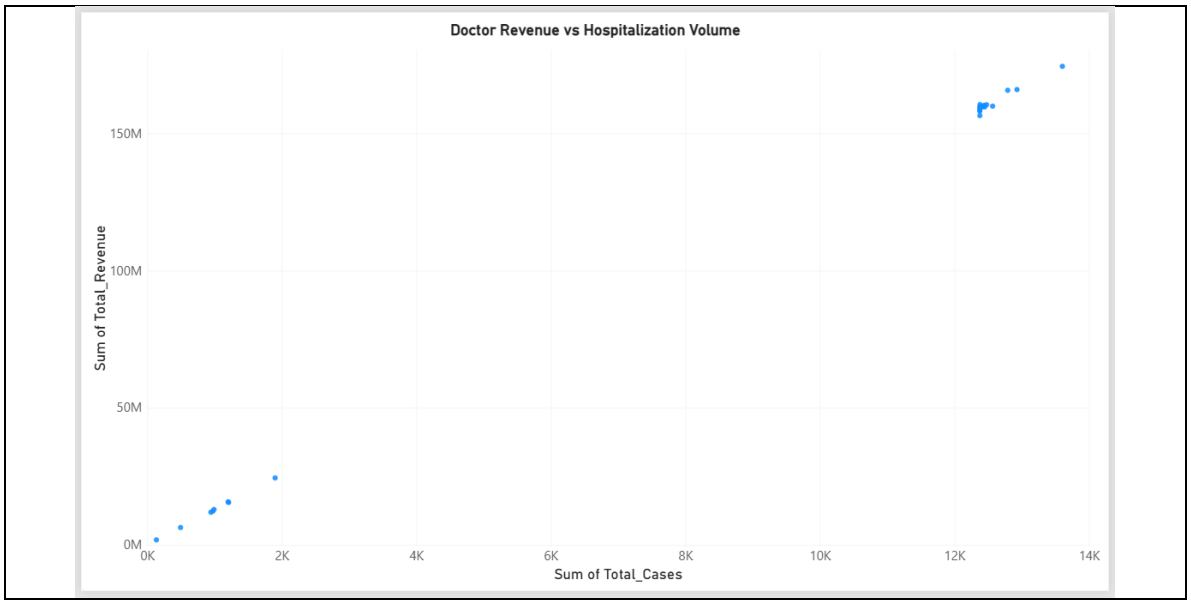


Fig 17: *Doctor Revenue vs Hospitalization Volume Visual*

Business Value: Useful for workforce planning, workload analysis, capacity management and incentive analysis.

Query 4: Which DRGs have high patient volume but relatively low revenue per case?

Question: Which treatments have high demand but comparatively low revenue per hospitalization?

```sql
/*
4. Which DRGs have high patient volume but relatively low revenue per case?
Question: Which treatments have high demand but comparatively low revenue per hospitalization?
*/
SELECT
    drg.DRG_Code,
    drg.DRG_Name,
    COUNT(*) AS Total_Cases,
    SUM(CAST(f.Net_Amount AS DECIMAL(18,2))) AS Total_Revenue,
    AVG(CAST(f.Net_Amount AS DECIMAL(18,2))) AS Avg_Revenue_Per_Case
FROM Fact_Hospitalization f
INNER JOIN Dim_DRG drg
    ON f.DRG_Code = drg.DRG_Code
GROUP BY
    drg.DRG_Code,
    drg.DRG_Name
HAVING COUNT(*) >= 100;
```

98 % 5 0

Results Messages

	DRG_Code	DRG_Name	Total_Cases	Total_Revenue	Avg_Revenue_Per_Case
1	3	ECMO OR TRACH W MV >96 HRS OR PDX EXC FACE, MOUTH ...	421	5551497.76	13186.455486
2	4	TRACH W MV >96 HRS OR PDX EXC FACE, MOUTH & NECK W/...	431	5483121.02	12721.858515
3	23	CRANIOTOMY W MAJOR DEVICE IMPLANT OR ACUTE CNS PDX...	297	3908382.78	13159.537979
4	25	CRANIOTOMY & ENDOVASCULAR INTRACRANIAL PROCEDURE...	549	7203581.15	13121.277140
5	26	CRANIOTOMY & ENDOVASCULAR INTRACRANIAL PROCEDURE...	255	3472167.93	13616.344823
6	27	CRANIOTOMY & ENDOVASCULAR INTRACRANIAL PROCEDURE...	267	3280031.28	12284.761348
7	38	EXTRACRANIAL PROCEDURES W CC	414	5399454.45	13042.160507
8	39	EXTRACRANIAL PROCEDURES W/O CC/MCC	841	11054416.94	13144.372104
9	41	PERIPH/CRANIAL NERVE & OTHER NERV SYST PROC W CC OR...	115	1461229.66	12706.344869
10	54	NERVOUS SYSTEM NEOPLASMS W MCC	391	4764878.93	12186.391125
11	56	DEGENERATIVE NERVOUS SYSTEM DISORDERS W MCC	321	3825372.49	11917.048255
12	57	DEGENERATIVE NERVOUS SYSTEM DISORDERS W/O MCC	885	11495523.14	12989.291683

Fig 18: *DRG Volume vs Revenue per Case Query*

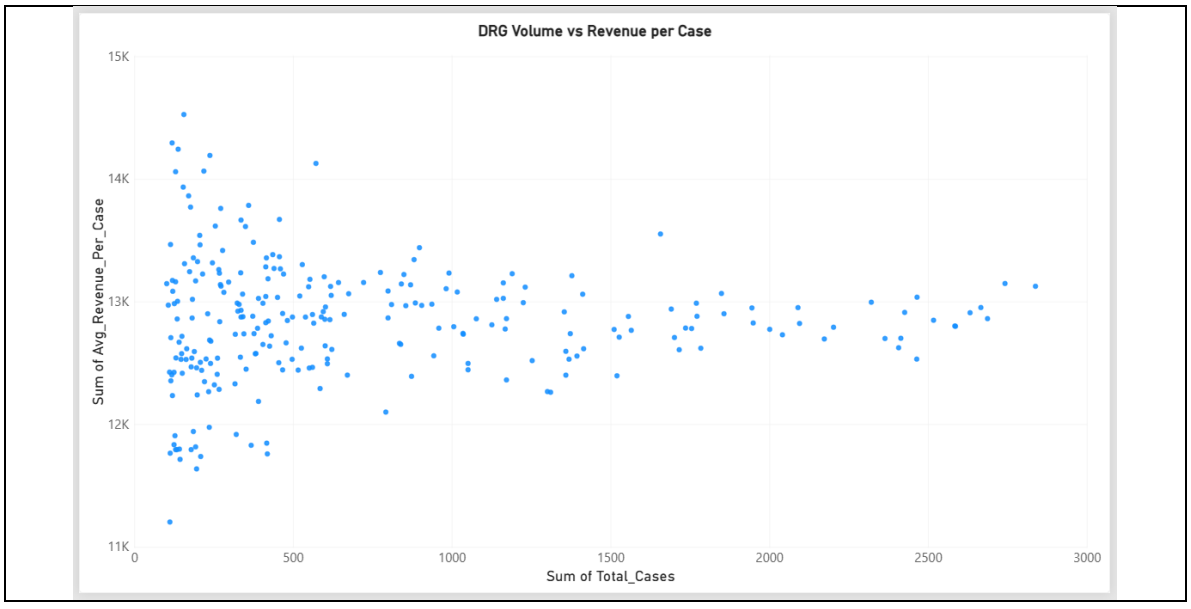


Fig 19: *DRG Volume vs Revenue per Case Visual*

Business Value: A high-volume/low-revenue service may require management to investigate pricing, reimbursement, operational cost or resource utilization.

Query 5: What is the month-over-month revenue growth trend?

Question: How is hospital revenue changing month-over-month, and which months experienced the strongest growth or decline?

```sql
/*
5. What is the month-over-month revenue growth trend?
Question: How is hospital revenue changing month-over-month, and which months experienced the strongest growth or decline?
*/
WITH MonthlyRevenue AS
(
    SELECT
        Year,
        Month,
        SUM(CAST(Net_Amount AS DECIMAL(18,2))) AS Total_Revenue
    FROM Fact_Hospitalization
    GROUP BY
        Year,
        Month
),
RevenueWithPrevious AS
(
    SELECT
        Year,
        Month,
        Total_Revenue,
        LAG(Total_Revenue) OVER (
            ORDER BY Year, Month
        ) AS Previous_Month_Revenue
    FROM MonthlyRevenue
)
SELECT
    Year,
    Month,
    Total_Revenue,
    Previous_Month_Revenue,
    Total_Revenue - Previous_Month_Revenue AS Revenue_Change,
    CAST(
        (Total_Revenue - Previous_Month_Revenue) * 100.0
        / NULLIF(Previous_Month_Revenue, 0)
        AS DECIMAL(10,2)
    ) AS Growth_Percentage
FROM RevenueWithPrevious
ORDER BY
    Year,
    Month;
```

67 % 5 0

Results Messages

	Year	Month	Total_Revenue	Previous_Month_Revenue	Revenue_Change	Growth_Percentage
1	2018	8	35852227.68	NULL	NULL	NULL
2	2018	9	43161313.63	35852227.68	7309085.95	20.39
3	2018	10	42231021.39	43161313.63	-930292.24	-2.16
4	2018	11	41256041.10	42231021.39	-974980.29	-2.31
5	2018	12	43281801.63	41256041.10	2025760.53	4.91
6	2019	1	43962507.74	43281801.63	680706.11	1.57

Fig 20: *Monthly Hospital Revenue Trend Query*

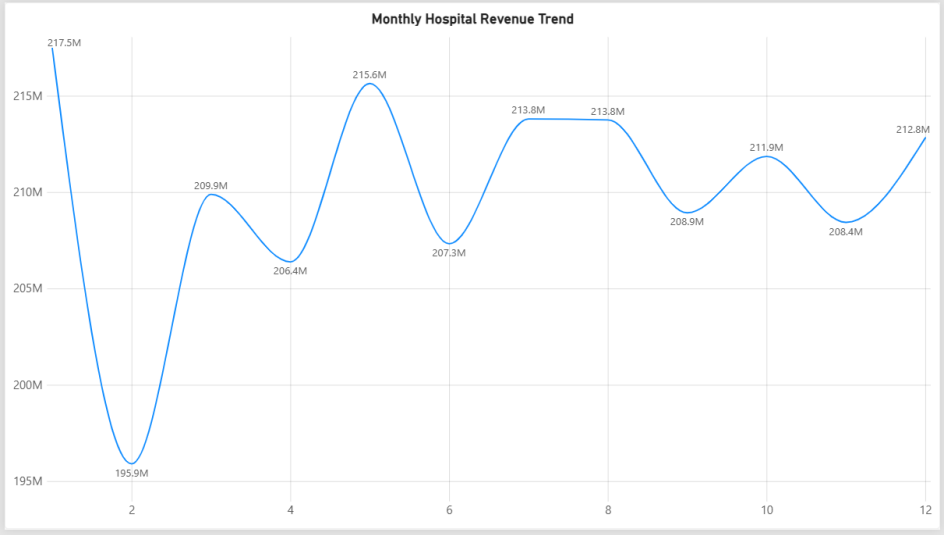


Fig 21: *Monthly Hospital Revenue Trend Visual*

Business Value: Helps Management can identify growth periods, declining periods and seasonal demand patterns for staffing and resource planning.

Query 6: Which doctors have unusually high or low cure rates?

Question: Which doctors have the highest and lowest observed cure rates?

```sql
/*
6. Which doctors have unusually high or low cure rates?
Which doctors have the highest and lowest observed cure rates?
*/
SELECT
    doc.Doctor_Name,
    COUNT(*) AS Total_Cases,
    SUM(
        CASE
            WHEN f.Has_Cured = 'YES' THEN 1
            ELSE 0
        END
    ) AS Cured_Cases,
    CAST(
        SUM(
            CASE
                WHEN f.Has_Cured = 'YES' THEN 1
                ELSE 0
            END
        ) * 100.0 / COUNT(*)
        AS DECIMAL(10,2)
    ) AS Cure_Rate
FROM Fact_Hospitalization f
INNER JOIN Dim_Doctor doc
    ON f.Doctor_Code_ID = doc.Doctor_Code_ID
GROUP BY
    doc.Doctor_Name
HAVING COUNT(*) >= 100
ORDER BY
    Cure_Rate DESC;
```

81 % ⊗ 5 ⚠ 0

Results | Messages

	Doctor_Name	Total_Cases	Cured_Cases	Cure_Rate
1	Surendra	12381	9028	72.92
2	Sagun	12448	9076	72.91
3	Manik	12375	9017	72.86
4	Shyam	12379	9007	72.76
5	Anish	12566	9126	72.62
6	Liza	12424	9022	72.62
7	Hari	12476	9031	72.39
8	Praful	12376	8959	72.39

Fig 22: *Revenue by Doctor and Month Query*

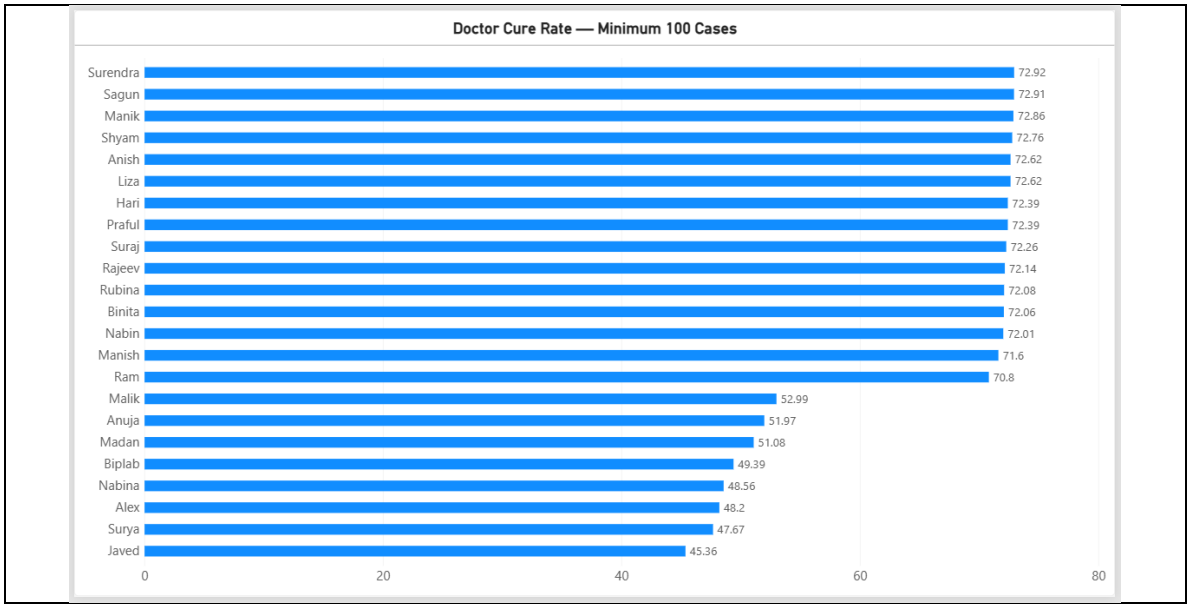


Fig 23: *Revenue by Doctor and Month PowerBI Visual*

Business Value: Useful as a quality-monitoring indicator, but it should not be interpreted as proof that one doctor provides better care because diagnosis severity and patient mix can differ.

Query 7: How does length of stay affect revenue and operational demand?

Question: How do short-, medium- and long-stay hospitalizations differ in volume and revenue?

```sql
/*
7. How does length of stay affect revenue and operational demand?
How do short-, medium- and long-stay hospitalizations differ in volume and revenue?
*/
SELECT
    CASE
        WHEN p.Admitted_Days BETWEEN 1 AND 3
            THEN 'Short Stay (1-3 Days)'
        WHEN p.Admitted_Days BETWEEN 4 AND 7
            THEN 'Medium Stay (4-7 Days)'
        WHEN p.Admitted_Days BETWEEN 8 AND 14
            THEN 'Long Stay (8-14 Days)'
        ELSE 'Very Long Stay (15+ Days)'
    END AS Stay_Category,
    COUNT(*) AS Total_Hospitalizations,
    SUM(CAST(f.Net_Amount AS DECIMAL(18,2))) AS Total_Revenue,
    AVG(CAST(f.Net_Amount AS DECIMAL(18,2))) AS Avg_Revenue_Per_Case,
    AVG(CAST(p.Admitted_Days AS DECIMAL(10,2))) AS Avg_Length_of_Stay
FROM Fact_Hospitalization f
INNER JOIN Dim_Patient p
    ON f.Patient_Code_ID = p.Patient_Code_ID
GROUP BY
    CASE
        WHEN p.Admitted_Days BETWEEN 1 AND 3
            THEN 'Short Stay (1-3 Days)'
        WHEN p.Admitted_Days BETWEEN 4 AND 7
            THEN 'Medium Stay (4-7 Days)'
        WHEN p.Admitted_Days BETWEEN 8 AND 14
            THEN 'Long Stay (8-14 Days)'
        ELSE 'Very Long Stay (15+ Days)'
    END
ORDER BY
    MIN(p.Admitted_Days);
/*
```

74 % 5 0

Results Messages

	Stay_Category	Total_Hospitalizations	Total_Revenue	Avg_Revenue_Per_Case	Avg_Length_of_Stay
1	Short Stay (1-3 Days)	21540	39834636.41	1849.333166	2.007288
2	Medium Stay (4-7 Days)	28983	145841425.67	5031.964450	5.497395
3	Long Stay (8-14 Days)	51215	516013344.53	10075.433848	10.999238
4	Very Long Stay (15+ Days)	94587	1820546453.01	19247.322073	20.987059

Fig 24: *Hospitalization Volume and Revenue by Length of Stay Query*

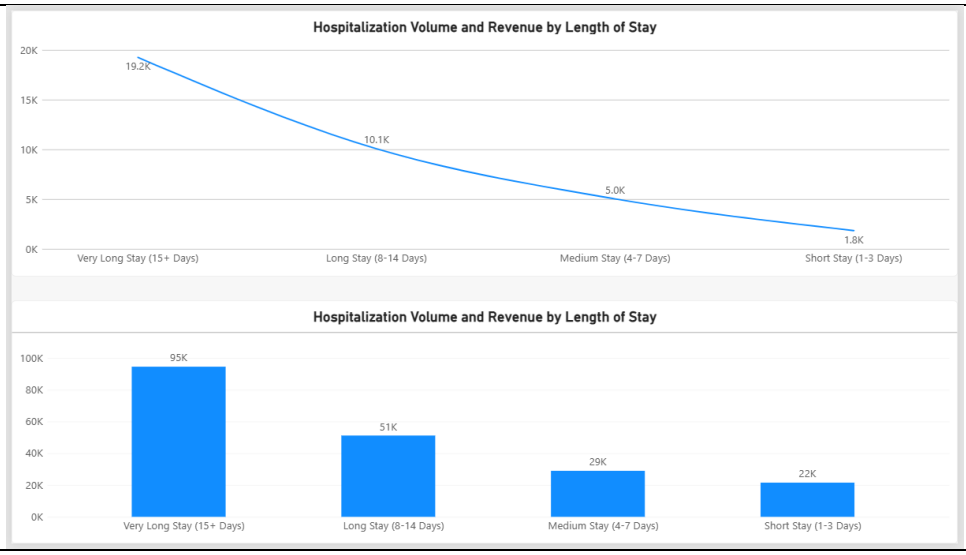


Fig 25: *Hospitalization Volume and Revenue by Length of Stay Visual*

Business Value: Helps with bed capacity, staffing, operational planning and long-stay management.

Query 8: Which DRGs have both high revenue and high mortality?

Question: Which major treatment categories require management attention because they combine substantial financial activity with elevated mortality?

```sql
/*
8. Which DRGs have both high revenue and high mortality?
Which major treatment categories require management attention because they combine substantial financial activity with elevated mortality?
*/
WITH DRG_Performance AS
(
    SELECT
        drg.DRG_Code,
        drg.DRG_Name,
        COUNT(*) AS Total_Cases,
        SUM(CAST(f.Net_Amount AS DECIMAL(18,2))) AS Total_Revenue,
        SUM(CASE WHEN f.Is_Dead = 'YES' THEN 1 ELSE 0 END) AS Deaths,
        CAST(SUM(CASE WHEN f.Is_Dead = 'YES' THEN 1 ELSE 0 END) * 100.0 / COUNT(*) AS DECIMAL(10,2)) AS Mortality_Rate
    FROM Fact_Hospitalization f
    INNER JOIN Dim_DRG drg
        ON f.DRG_Code = drg.DRG_Code
    GROUP BY
        drg.DRG_Code,
        drg.DRG_Name
    HAVING COUNT(*) >= 100
)
SELECT
    DRG_Code,
    DRG_Name,
    Total_Cases,
    Total_Revenue,
    Deaths,
    Mortality_Rate
FROM DRG_Performance;
```

74 % 5 0

Results Messages

	DRG_Code	DRG_Name	Total_Cases	Total_Revenue	Deaths	Mortality_Rate
1	3	ECMO OR TRACH W MV >96 HRS OR PDX EXC FACE, MOUTH ...	421	5551497.76	144	34.20
2	4	TRACH W MV >96 HRS OR PDX EXC FACE, MOUTH & NECK W/...	431	5483121.02	152	35.27
3	23	CRANIOTOMY W MAJOR DEVICE IMPLANT OR ACUTE CNS PDX...	297	3908382.78	100	33.67
4	25	CRANIOTOMY & ENDOVASCULAR INTRACRANIAL PROCEDURE...	549	7203581.15	187	34.06
5	26	CRANIOTOMY & ENDOVASCULAR INTRACRANIAL PROCEDURE...	255	3472167.93	82	32.16
6	27	CRANIOTOMY & ENDOVASCULAR INTRACRANIAL PROCEDURE...	267	3280031.28	86	32.21
7	38	EXTRACRANIAL PROCEDURES W CC	414	5399454.45	126	30.43

Fig 26: *DRG Revenue vs Mortality Query*

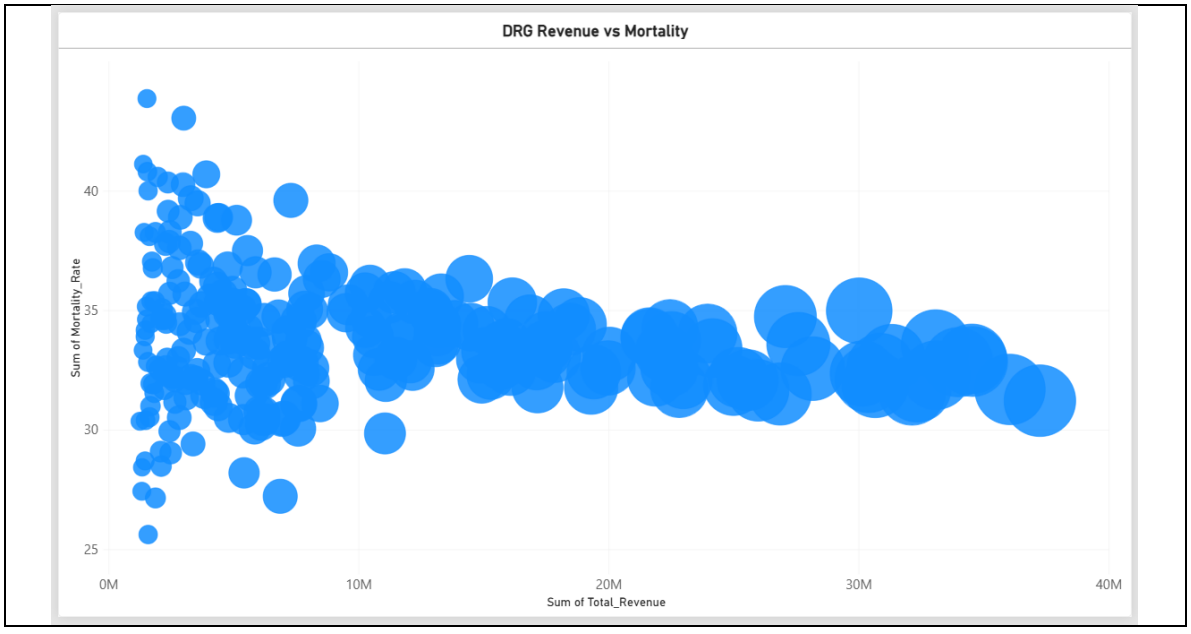


Fig 27: *DRG Revenue vs Mortality Visual*

Business Value: This identifies areas where financial importance and clinical risk intersect.

Query 9: What percentage of gross amount is represented by discounts?

Question: How much revenue is being reduced through patient discounts, and what is the effective discount rate?

```sql
/*
9. What percentage of gross revenue is being lost through discounts?
or, What percentage of gross amount is represented by discounts?
How much revenue is being reduced through patient discounts, and what is the effective discount rate?
*/
SELECT
    SUM(CAST(G_Amount AS DECIMAL(18,2))) AS Total_Gross_Amount,
    SUM(CAST(D_Amt AS DECIMAL(18,2))) AS Total_Discount,
    SUM(CAST(Net_Amount AS DECIMAL(18,2))) AS Total_Net_Revenue,
    CAST(SUM(CAST(D_Amt AS DECIMAL(18,2))) * 100.0 / NULLIF(SUM(CAST(G_Amount AS DECIMAL(18,2))), 0) AS DECIMAL(10,2)
    ) AS Effective_Discount_Percentage
FROM Fact_Hospitalization;

/*
```

89 % 5 0

Results Messages

	Total_Gross_Amount	Total_Discount	Total_Net_Revenue	Effective_Discount_Percentage
1	2480035417.00	247968318.57	2522235859.62	10.00

Fig 28: *Gross amount, discounts and net revenue Query*



Fig 29: *Gross amount, discounts and net revenue Visual*

Business Value: Supports discount-policy evaluation, revenue optimization and financial reporting.

Query 10: Which district–DRG combinations have the highest hospitalization demand?

Question: Which combinations of geographic district and treatment category generate the highest hospitalization volume?

```sql
/*
10. Which district-DRG combinations have the highest hospitalization demand?
Which combinations of geographic district and treatment category generate the highest hospitalization volume?
*/
SELECT TOP 10
    d.District_Name,
    drg.DRG_Name,
    COUNT(*) AS Total_Hospitalizations,
    SUM(CAST(f.Net_Amount AS DECIMAL(18,2))) AS Total_Revenue
FROM Fact_Hospitalization f
INNER JOIN Dim_District d
    ON f.District_Code_ID = d.District_Code_ID
INNER JOIN Dim_DRG drg
    ON f.DRG_Code = drg.DRG_Code
GROUP BY
    d.District_Name,
    drg.DRG_Name
ORDER BY
    Total_Hospitalizations DESC;

-- or
SELECT TOP 10
    CONCAT(d.District_Name, ' – ', drg.DRG_Name) AS District_DRG,
    d.District_Name,
    drg.DRG_Name,
    COUNT(*) AS Total_Hospitalizations,
    SUM(CAST(f.Net_Amount AS DECIMAL(18,2))) AS Total_Revenue
FROM Fact_Hospitalization f
INNER JOIN Dim_District d
    ON f.District_Code_ID = d.District_Code_ID
INNER JOIN Dim_DRG drg
    ON f.DRG_Code = drg.DRG_Code
GROUP BY
    d.District_Name,
    drg.DRG_Name
ORDER BY
    Total_Hospitalizations DESC;
```

74 % ⊗ 5 ⚠ 0

Results | Messages

	District_Name	DRG_Name	Total_Hospitalizations	Total_Revenue
1	Maine	CELLULITIS W/O MCC	71	970621.64
2	Maine	SEPTICEMIA OR SEVERE SEPSIS W/O MV >96 HOURS W MCC	71	944176.01
3	Wisconsin	CHRONIC OBSTRUCTIVE PULMONARY DISEASE W MCC	70	908376.34
4	Tennessee	KIDNEY & URINARY TRACT INFECTIONS W/O MCC	70	890348.87
5	WestVirginia	SEPTICEMIA OR SEVERE SEPSIS W/O MV >96 HOURS W MCC	69	865169.46
6	Vermont	CHRONIC OBSTRUCTIVE PULMONARY DISEASE W MCC	68	813499.30

Fig 30: *Top 10 District-DRG combinations by Hospitalization Demand Query*

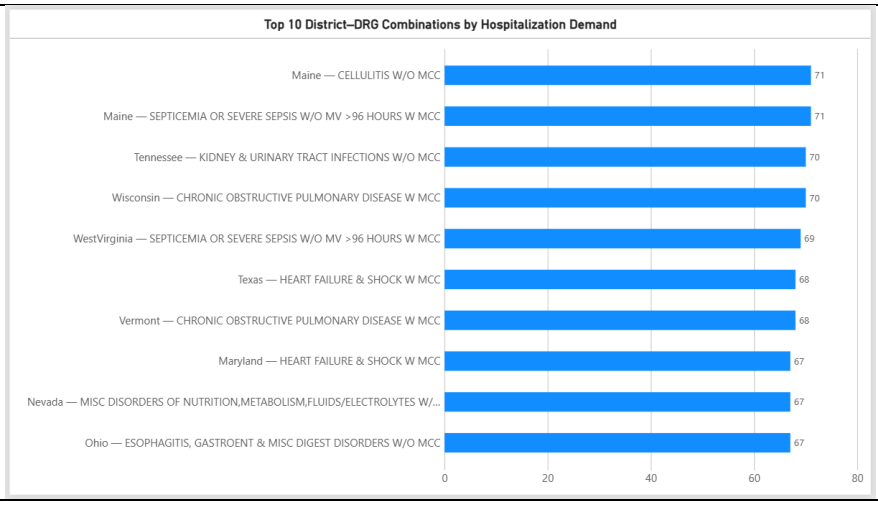


Fig 31: *Top 10 District-DRG combinations by Hospitalization Demand Visual*

Business Value: Management can identify where specific treatments have the highest demand, which can support specialist allocation and regional service planning.

14. Data Quality & Validation Rules

The project includes multi-level checks:

- Null validation / Missing value checks
- Date validation
- Numeric field verification
- Standardization (whitespace trim)

15. Advantages of ETL Process

- Provides a single source of truth by integrating raw data into a clean fact table.
- Ensures data quality via transformation and validation.
- Improves query performance by standardizing data types and indexing.
- Enables business insights through advanced analytics queries.

16. Disadvantages / Challenges

- High development time for writing and maintaining SQL scripts.
- Scalability issues if dataset grows beyond millions of records.
- Bulk insert requires strict file format consistency (delimiters, headers).
- Potential data loss if invalid rows are not handled properly during transformation.

17. Acknowledgment

This project is based on the Medicare Inpatient 2017 dataset. The ETL process was implemented using Microsoft SQL Server and SQL Server Management Studio (SSMS).

18. Conclusion

This ETL project demonstrates a robust data engineering strategy for transforming raw patient data into a well-structured analytical warehouse. The layered architecture ensures traceability, accuracy, and high performance. The analytical queries provide actionable insights into hospital operations and patient care.

19. Glossary

The glossary defines key technical terms used throughout the ETL documentation. It includes definitions for ETL concepts such as “staging layer,” “data warehouse,” and “fact table,” along with industry-specific terms such as DRG, mortality indicator, VAT, and net amount. Providing these definitions ensures clarity for readers from both technical and non-technical backgrounds.

Such as:

- **ETL:** Extract, Transform, Load
- **DRG:** Diagnosis Related Group
- **DWH:** Data Warehouse
- **OLAP:** Online Analytical Processing
- **Raw Layer:** Ingestion layer holding unprocessed data
- **Staging:** Temporary processing layer
- **Fact Table:** Table containing measurable business data
- **Dimension Table:** Table describing attributes/context of facts
- **VAT:** Value Added Tax